Modern Models, Historical isiXhosa: A Held-Out Study of Language-Modelling Fit

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Abstract

Language coverage does not establish coverage of a language's historical varieties. We evaluate five frozen pretrained models on nineteenth-century isiXhosa newspapers and a modern NCHLT comparison corpus. Final evaluation uses 300 sentences per corpus from disjoint newspaper issues and modern documents, with character-length stratification. XLM-R, Afro-XLMR and InkubaLM assign worse character-normalised scores to historical text, with median gaps of 0.796, 0.815 and 0.627 and source-group bootstrap intervals excluding zero. mBERT and AfriBERTa show no clear directional difference. Validation reproduces the same pattern. These findings indicate a historical-modern corpus mismatch for three models; they do not demonstrate semantic failure or isolate temporal change from register, orthography and transcription differences.

Introduction

The IsiXhosa Intellectual Traditions Digital Archive preserves writing that is central to studying African intellectual history. Making this material digitally accessible, however, does not establish that present-day NLP systems process it adequately. We ask: do selected modern pretrained models assign worse language-modelling scores to isiXhosa newspaper text from 1870-1894 than to a modern comparison corpus? We hypothesise a positive historical-modern scoring gap in models with documented isiXhosa training coverage. Historical English research finds benefits from historically specific pretraining, motivating evaluation across periods rather than assuming transfer. African-language adaptation likewise improves selected downstream benchmarks, while AfriBERTa demonstrates the value of targeted multilingual pretraining. These studies motivate our experiment but do not establish historical isiXhosa performance. We use pseudo-log-likelihood (PLL) for masked models [5] and causal log-likelihood for InkubaLM. The experiment measures distributional fit, not semantic understanding or whether a model identifies the text as a foreign dialect.

Methods

The historical input comprises the supplied isiXhosa text blocks from Imvo Zabantsundu and Isigidimi sama-Xosa. The tracker covers 3,608 pages; a 100-isiXhosa-word minimum retains 3,592. Automated normalisation, segmentation and filtering yield 327,265 sentences from 3,574 pages. Sentences require at least three whitespace words and 25 characters, with at most 80 words. A proposed 90% page-level isiXhosa word-share filter was rejected: the diagnostic showed it would additionally exclude 895 substantial pages, although their English blocks were already separated. Figure S1 shows the diagnostic thresholds. The spike at 3,000 reflects display clipping, not an exact page word count. Modern comparison text is the NCHLT clean isiXhosa corpus [6], associated with text-resource development for South African languages [7]. After automated filtering, duplicate removal and a 200-sentence document cap, 41,188 sentences remain. Government-domain provenance [8] makes this a register-mismatched comparator. We performed no manual correction or modernisation because we lacked isiXhosa proficiency. Automated processing is distinct from manual linguistic verification. After exploratory scoring, we created source-disjoint development, validation and test partitions. All pages of a historical issue share a partition; modern grouping uses original document identifiers. Previously evaluated sources are assigned to development. Normalised exact duplicates and exact cross-corpus overlaps are removed. The resulting pools contain 263,748 historical and 41,188 modern sentences. Validation and test each contain 300 sentences per corpus, sampled equally across five character-length bins defined from development historical text (seed 20261031). Validation covers 100 historical issues and 114 modern documents; test covers 101 and 112. Models remain frozen: the development partition is not used for training or hyperparameter optimisation. The checkpoints are bert-base-multilingual-cased, castorini/afriberta_large, FacebookAI/xlm-roberta-large, Davlan/afro-xlmr-large and lelapa/InkubaLM-0.4B. XLM-R uses multilingual web pretraining [9]; the Afro-XLMR and InkubaLM documentation explicitly includes isiXhosa [10,11]. mBERT and AfriBERTa serve as comparison baselines, without isiXhosa in their documented language lists [4,12]. For each sentence, we divide total negative log score by its character count. Masked models score each non-special subword while other tokens remain visible; the causal model uses shifted next-token loss. The primary statistic is the historical minus modern median within each model. We report 95% percentile intervals from 2,000 source-group bootstrap draws, Cliff's delta, and approximate centred-bootstrap p-values with Holm correction across the five models

Results

The held-out test reproduces validation (Table 1; Figures S2-S3). All three positive-gap model intervals exclude zero; approximate Holm-adjusted bootstrap p-values are 0.0025. Both baseline intervals include zero. Test mean lengths are closely aligned: 111.41 historical versus 111.47 modern characters (medians 91 and 100). Figure S4 documents the original exploratory comparison. [1]

[1] Code and results: https://github.com/lisabeeaan/NLP-Project2

Table 1: Historical minus modern median negative (pseudo) log score per character. Positive gaps indicate worse historical fit within a model. Test: 300 sentences per corpus. Intervals resample source groups; delta is a descriptive sentence-order effect. Absolute scores and gap magnitudes are not a common performance ranking across masked and causal objectives.

Model	Validation gap	Test gap (95% CI)
mBERT	-0.039	79.88 [-0.128, 0.024]
AfriBERTa	-0.060	62.50 [-0.067, 0.020]
XLM-R	0.857	84.50 [0.696, 0.872]
Afro-XLMR	0.842	62.50 [0.725, 0.885]
InkubaLM	0.629	0.627 [0.578, 0.682]

Discussion

The stable direction across validation and test supports a corpus-level historical disadvantage for XLM-R, Afro-XLMR and InkubaLM under the chosen objectives. It does not show that these models cannot represent archival isiXhosa meaningfully. Nor does the baseline pattern establish historical robustness: a small difference can arise when neither corpus is well modelled. The small AfriBERTa reverse difference from the initial exploration is not clearly reproduced in either held-out partition. Language inclusion alone therefore should not be treated as sufficient evidence of historical coverage, and these five checkpoints do not identify a causal effect of pretraining language composition. Earlier exploratory comparisons found higher scores, hence poorer fit, for Imvo than Isigidimi in all five models. This is a newspaper contrast, not evidence that progressively older language is harder: publication period, content and text quality overlap. Similarly, NCHLT government text and historical newspapers differ in register and topic. Orthographic conventions, residual errors, unknown pretraining overlap and imperfect automated sentence boundaries could contribute to the main gap. Character normalisation reduces dependence on token counts but does not eliminate tokenizer effects. Standard subword PLL can inflate scores when other pieces of the target word remain visible [13]. The retained objective constrains interpretation. AfriBERTa also shows metric sensitivity: its secondary per-token test gap is 0.223 [0.045, 0.361], unlike its near-zero primary per-character gap. An exploratory synthetic-noise check on 100 modern sentences found that nominal corruption of 1%, 2% and 5% of letters produced score shifts equivalent to approximately 18-24%, 38-46% and 79-87% of the original mean per-token gaps in the three positive-gap models. These are sensitivity comparisons under a generic corruption mechanism, not measured percentages of historical degradation caused by OCR. They use the exploratory sample and a different summary statistic from Table 1. Actual residual transcription error and its score effect remain unmeasured without scan-verified reference passages. For responsible deployment, archival coverage warrants direct evaluation rather than inference from a language label. Poorer modelling fit could affect archival retrieval, classification or other access tools, but no downstream harm was measured here. Researchers, heritage institutions and isiXhosa-speaking communities could be affected if such systems privilege contemporary forms. Conversely, aggressively removing unusual forms or modernising spelling without expertise risks discarding historical evidence. Appropriate next steps are speaker- and historian-informed evaluation, register-matched modern comparisons, and verified transcription audits. Historical-domain adaptation is a research direction suggested by prior work, not an intervention tested in this project.

Conclusion

Three evaluated models assign consistently worse language-modelling scores to the sampled nineteenth-century isiXhosa corpus than to modern NCHLT text, including on a source-disjoint final test. The result supports a historical-modern corpus mismatch while leaving its linguistic and technical causes unresolved. It motivates historically informed evaluation; it does not establish semantic failure, dialect classification or measured downstream exclusion.

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